

Omer Ebead

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SUMMARY

AI researcher focused on multi-agent reinforcement learning and AI safety, with research experience at InstaDeep and the Cooperative AI Foundation. Author of a paper accepted at NeurIPS 2026 and co-first author of research on offline MARL generalisation, with experience in Python, JAX and full-stack development.

EDUCATION

Stellenbosch University | AIMS South Africa

Master's in Mathematical Sciences
AI for Science programme

Sep. 2024 to Jul. 2025
Cape Town, South Africa

University of Khartoum

Bachelor of Science (Honours) in Electrical and Electronic Engineering

Aug. 2016 to May 2021
Khartoum, Sudan

EXPERIENCE

Cooperative AI Foundation | Python, LLMs, Concordia

Research Fellow

Feb. 2026 to present
Cape Town, South Africa

- Built a Python simulation pipeline with Concordia to study how social context shapes information sharing between LLM agents.
- Authored research on contextual integrity in multi-agent LLM systems, accepted at NeurIPS 2026 and the ICML 2026 AI for Good Workshop.

InstaDeep | Python, JAX

Research Engineer Intern

Aug. 2025 to Jan. 2026
Cape Town, South Africa

- Extended multi-task sequence models in Python and JAX, prepared offline datasets, and conducted experiments evaluating generalisation to unseen multi-agent tasks.
- Co-first-authored research on offline multi-agent reinforcement learning, contributing theoretical analysis and manuscript writing.

Freelancer | Flask, Django, Next.js

Software Engineer

Apr. 2023 to Sep. 2024
Riyadh, Saudi Arabia

- Built React and Next.js interfaces for an online clinic appointment website (Clinical) and a washing company dashboard (Rewash).

AmunData | React, Flask, Power BI

Software Engineer

Apr. 2022 to Apr. 2023
Khartoum, Sudan

- Developed web applications with React frontends, Flask backends and database integration.
- Processed satellite imagery to calculate vegetation indices, including NDVI, VHI and VCI, and built models to forecast their future values.
- Collaborated with data analysts to create Power BI visualisations.

OROOMA | React, Kotlin

Software Engineer

Jan. 2021 to Apr. 2022
Khartoum, Sudan

- Developed web interfaces and backend services for two projects: a shipping company dashboard and the Suhub platform.

PUBLICATIONS

Omer Ebead, et al. (2026). *A Generative Model of Contextual Integrity: Appropriate vs. Inappropriate Sharing*. Accepted at NeurIPS 2026 and the ICML 2026 AI for Good Workshop.

Oussama Hidaoui*, **Omer Ebead***, et al. (2026). *Out-of-Distribution Generalisation with Sequence Models in Offline Multi-Agent Reinforcement Learning*. arXiv:2609.03667.

*Equal contribution.

PROJECTS

- Multi-Task Multi-Agent Reinforcement Learning** May 2025 to Jul. 2025
Python, JAX, Flax
- Extended the Sable network architecture to support multi-task, multi-environment reinforcement learning using Python, JAX and Flax.
- Recommender System** Nov. 2024
Python
- Implemented ALS-based collaborative filtering on MovieLens using Python and NumPy, with user/item biases and latent factors.
 - Built a Next.js frontend and Flask API to generate movie recommendations from user-provided movie ratings.
- Temporal Analysis of Regional Sustainability Using CNNs and Satellite Data** May 2020
Python, PyTorch
- Trained and evaluated CNNs in PyTorch for land-cover classification on the EuroSAT dataset.
 - Estimated biocapacity for Tuti Island and Port Sudan using predicted land-cover areas, yield factors and equivalence factors.

AWARDS

- Academic Excellence Award, AI for Science Master’s** Aug. 2025
Awarded for overall performance across academic coursework, thesis and graduation project.
- Google DeepMind Scholarship** Sep. 2024
Fully funded master’s scholarship at AIMS South Africa.

TECHNICAL SKILLS

Programming: Python, SQL (PostgreSQL), JavaScript, Java, C, C++
Machine Learning: JAX, Flax, PyTorch, TensorFlow, Keras
Data Analysis: NumPy, Pandas, Matplotlib, Power BI
Web Development: FastAPI, Flask, Django, React, Next.js, Node.js, HTML/CSS
Development Tools and Cloud: Git, Docker, Azure, Google Cloud

LANGUAGES AND PROFESSIONAL DEVELOPMENT

Languages: Arabic (native); English (IELTS 7.0).
Design Thinking: Hasso Plattner d-school Afrika.
Leadership: McKinsey Forward Program.